

Question 01

$$A = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{bmatrix} = \begin{bmatrix} 1 & -2 & 3 \\ 6 & 7 & -1 \\ -3 & 1 & 4 \end{bmatrix}$$

Find all minors & cofactors.

$$m_{11} = \begin{vmatrix} 7 & -1 \\ 1 & 4 \end{vmatrix} = 28 + 1 = 29$$

$$m_{12} = \begin{vmatrix} 6 & -1 \\ -3 & 4 \end{vmatrix} = 24 - 3 = 21$$

$$m_{13} = \begin{vmatrix} 6 & 7 \\ -3 & 1 \end{vmatrix} = 6 + 21 = 27$$

$$m_{21} = \begin{vmatrix} -2 & 3 \\ 1 & 4 \end{vmatrix} = -8 - 3 = -11$$

$$m_{22} = \begin{vmatrix} 1 & 3 \\ -3 & 4 \end{vmatrix} = 4 + 9 = 13$$

$$m_{23} = \begin{vmatrix} 1 & -2 \\ -3 & 1 \end{vmatrix} = 1 - 6 = -5$$

$$m_{31} = \begin{vmatrix} -2 & 3 \\ 7 & -1 \end{vmatrix} = 2 - 21 = -19$$

$$m_{32} = \begin{vmatrix} 1 & 3 \\ 6 & -1 \end{vmatrix} = -1 - 18 = -19$$

$$m_{33} = \begin{vmatrix} 1 & -2 \\ 6 & 7 \end{vmatrix} = 7 + 12 = 19$$

$$C_{11} = (-1)^{1+1}(29) = 29$$

$$C_{12} = (-1)^{1+2}(21) = -21$$

$$C_{13} = (-1)^{1+3}(+24) = 24$$

$$C_{21} = (-1)^{2+1}(-11) = 11$$

$$C_{22} = (-1)^{2+2}(13) = 13$$

$$C_{23} = (-1)^{2+3}(-5) = 5$$

$$C_{31} = (-1)^{3+1}(-19) = -19$$

$$C_{32} = (-1)^{3+2}(-19) = 19$$

$$C_{33} = (-1)^{3+3}(19) = 19$$

$$2- A = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 3 & 6 \\ 0 & 1 & 4 \end{bmatrix}$$

$$M_{11} = \begin{vmatrix} 3 & 6 \\ 1 & 4 \end{vmatrix} = 12 - 6 = 6$$

$$C_{11} = (-1)^{1+1}(6) = 6$$

$$M_{12} = \begin{vmatrix} 3 & 6 \\ 0 & 4 \end{vmatrix} = 12 - 0 = 12$$

$$C_{12} = (-1)^{1+2}(12) = -12$$

$$M_{13} = \begin{vmatrix} 3 & 3 \\ 0 & 1 \end{vmatrix} = 3 - 0 = 3$$

$$C_{13} = (-1)^{1+3}(3) = 3$$

$$M_{21} = \begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} = 4 - 2 = 2$$

$$C_{21} = (-1)^{1+2} (2) = -2$$

$$M_{22} = \begin{vmatrix} 1 & 2 \\ 0 & 4 \end{vmatrix} = 4 - 0 = 4$$

$$C_{22} = (-1)^{2+2} (4) = 4$$

$$M_{23} = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1 - 0 = 1$$

$$C_{23} = (-1)^{2+3} (1) = -1$$

$$M_{31} = \begin{vmatrix} 1 & 2 \\ 3 & 6 \end{vmatrix} = 6 - 6 = 0$$

$$C_{31} = 0$$

$$M_{32} = \begin{vmatrix} 1 & 2 \\ 3 & 6 \end{vmatrix} = 6 - 6 = 0$$

$$C_{32} = 0$$

$$M_{33} = \begin{vmatrix} 1 & 1 \\ 3 & 3 \end{vmatrix} = 3 - 3 = 0$$

$$C_{33} = 0$$

$$3- A = \begin{bmatrix} 4 & -1 & 1 & 6 \\ 0 & 0 & -3 & 3 \\ 4 & 1 & 0 & 14 \\ 4 & 1 & 3 & 2 \end{bmatrix}$$

m_{13} and C_{13}

$$m_{13} = \begin{vmatrix} 0 & 0 & 3 \\ 4 & 1 & 14 \\ 4 & 1 & 2 \end{vmatrix} \rightarrow 3 \begin{vmatrix} 4 & 1 \\ 4 & 1 \end{vmatrix} = 0$$

$$C_{13} = 0$$

M_{23} and C_{23}

$$m_{23} = \begin{vmatrix} 4 & -1 & 6 \\ 4 & 1 & 14 \\ 4 & 1 & 2 \end{vmatrix}$$

$$= 4 \begin{vmatrix} 1 & 14 \\ 1 & 2 \end{vmatrix} - 1 \begin{vmatrix} 4 & 14 \\ 4 & 2 \end{vmatrix} + 6 \begin{vmatrix} 4 & 1 \\ 4 & 1 \end{vmatrix}$$

$$= 4(2-14) + 1(8-56) + 6(0)$$

$$= 4(-12) + 1(-48) = -48 - 48 = -96$$

$$C_{23} = (-1)^{2+3}(-96) = -96$$

M_{22} and C_{22}

$$m_{22} = \begin{vmatrix} 4 & 1 & 6 \\ 4 & 0 & 14 \\ 4 & 3 & 2 \end{vmatrix}$$

$$C_{22} = (-1)^{2+2}(-48) = 48$$

$$= 4 \begin{vmatrix} 0 & 14 \\ 3 & 2 \end{vmatrix} - 1 \begin{vmatrix} 4 & 14 \\ 4 & 2 \end{vmatrix} + 6 \begin{vmatrix} 4 & 0 \\ 4 & 3 \end{vmatrix} = 4(-42) - 1(8-56) + 6(12)$$

$$= -168 + 48 + 72 = -48$$

M_{21} and C_{21}

$$m_{21} = \begin{vmatrix} -1 & 1 & 6 \\ 1 & 0 & 14 \\ 1 & 3 & 2 \end{vmatrix}$$

$$= -1 \begin{vmatrix} 0 & 14 \\ 3 & 2 \end{vmatrix} - 1 \begin{vmatrix} 1 & 14 \\ 1 & 2 \end{vmatrix} + 6 \begin{vmatrix} 1 & 0 \\ 1 & 3 \end{vmatrix}$$

$$= -1(0 - 42) - 1(2 - 14) + 6(3 - 0)$$

$$= 42 + 12 + 18 = 72$$

$$C_{21} = (-1)^{2+1}(72) = -72$$

Question # 4

$$A = \begin{bmatrix} 2 & 3 & -1 & 1 \\ -3 & 2 & 0 & 3 \\ 3 & -2 & 1 & 0 \\ 3 & -2 & 1 & 4 \end{bmatrix}$$

 M_{32} and C_{32}

$$m_{32} = \begin{vmatrix} 2 & -1 & 1 \\ -3 & 0 & 3 \\ 3 & 1 & 4 \end{vmatrix}$$

$$= 2 \begin{vmatrix} 0 & 3 \\ 1 & 4 \end{vmatrix} + 1 \begin{vmatrix} -3 & 3 \\ 3 & 4 \end{vmatrix} + 1 \begin{vmatrix} -3 & 0 \\ 3 & 1 \end{vmatrix}$$

$$= 2(0 - 3) + 1(-12 - 9) + 1(-3 - 0)$$

$$= -6 - 21 - 3 = -30$$

$$C_{32} = (-1)^{3+2}(-30) = -30$$

M_{44}, C_{44}

$$M_{44} = \begin{vmatrix} 2 & 3 & -1 \\ -3 & 2 & 0 \\ 3 & -2 & 1 \end{vmatrix}$$

$$= 2 \begin{vmatrix} 2 & 0 \\ -2 & 1 \end{vmatrix} - 3 \begin{vmatrix} -3 & 0 \\ 3 & 1 \end{vmatrix} - 1 \begin{vmatrix} -3 & 2 \\ 3 & -2 \end{vmatrix}$$

$$= 2(2+0) - 3(-3+0) - 1(6-6)$$

$$= 4 + 9 - 0 = 13$$

$$C_{44} = (-1)^{4+4}(13) = 13$$

 M_{41}, C_{41}

$$M_{41} = \begin{vmatrix} 3 & -1 & 1 \\ 2 & 0 & 3 \\ -2 & 1 & 0 \end{vmatrix}$$

$$= 3 \begin{vmatrix} 0 & 3 \\ 1 & 0 \end{vmatrix} + 1 \begin{vmatrix} 2 & 3 \\ -2 & 0 \end{vmatrix} + 1 \begin{vmatrix} 2 & 0 \\ -2 & 1 \end{vmatrix}$$

$$= 3(0-3) + 1(0+6) + 1(2+0)$$

$$= -9 + 6 + 2 = -1$$

$$C_{41} = (-1)^{4+1}(-1) = 1$$

 M_{24}, C_{24}

$$M_{24} = \begin{vmatrix} 2 & 3 & -1 \\ 3 & -2 & 1 \\ 3 & -2 & 1 \end{vmatrix}$$

Determinant is 0

$$C_{24} = 0$$

Evaluate determinant of matrix
if matrix is invertible
then find inverse.

$$5 - \begin{bmatrix} 3 & 5 \\ -2 & 4 \end{bmatrix}$$

$$\text{Det}(A) = 12 + 10 = 22$$

$$A^{-1} = \frac{1}{22} \begin{bmatrix} 4 & -5 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 2/11 & -5/22 \\ 1/11 & 3/22 \end{bmatrix}$$

$$6 - \begin{bmatrix} 4 & 1 \\ 8 & 2 \end{bmatrix}$$

$$\text{Det}(A) = 8 - 8 = 0$$

Inverse not possible

$$7 - \begin{bmatrix} -5 & 7 \\ -7 & -2 \end{bmatrix}$$

$$\text{Det}(A) = 10 + 49 = 59$$

$$A^{-1} = \frac{1}{59} \begin{bmatrix} -2 & -7 \\ 7 & -5 \end{bmatrix} = \begin{bmatrix} -2/59 & -7/59 \\ 7/59 & -5/59 \end{bmatrix}$$

$$8 - A = \begin{bmatrix} \sqrt{2} & \sqrt{6} \\ 4 & \sqrt{3} \end{bmatrix}$$

$$\text{Det}(A) = \sqrt{6} - 4\sqrt{6} = -3\sqrt{6}$$

$$A^{-1} = \frac{1}{\begin{vmatrix} \sqrt{3} & -\sqrt{6} \\ -4 & \sqrt{2} \end{vmatrix}} = \frac{1}{-3\sqrt{6}} \begin{bmatrix} -3\sqrt{18} & 18 \\ 12\sqrt{6} & -3\sqrt{12} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sqrt{3}}{-3\sqrt{6}} & \frac{-\sqrt{6}}{-3\sqrt{6}} \\ \frac{-4}{-3\sqrt{6}} & \frac{\sqrt{2}}{-3\sqrt{6}} \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{18}} & \frac{1}{3} \\ \frac{4}{3\sqrt{6}} & -\frac{1}{3\sqrt{3}} \end{bmatrix}$$

Arrow technique to find
Determinant

$$9- \begin{vmatrix} a-3 & 5 \\ -3 & a-2 \end{vmatrix}$$

$$= (a-3)(a-2) - 5(-3)$$

$$= a(a-2) - 3(a-2) + 15$$

$$= a^2 - 2a - 3a + 6 + 15$$

$$= a^2 - 5a + 21$$

$$10- \begin{vmatrix} -2 & 7 & 6 \\ 5 & 1 & -2 \\ 3 & 8 & 4 \end{vmatrix}$$

$$\begin{vmatrix} -2 & 7 & 6 \\ 5 & 1 & -2 \\ 3 & 8 & 4 \end{vmatrix}$$

$$= -2(1)(4) + 7(-2)(3) + 6(5)(8) - 6(1)(3) - (-2)(-2)8 - 7(5)(4)$$

$$= -8 - 42 + 240 - 18 - 32 - 140 = 0$$

11-

$$\begin{vmatrix} -2 & 1 & 4 \\ 3 & 5 & -7 \\ 1 & 6 & 2 \end{vmatrix}$$

$$\begin{vmatrix} -2 & 1 & 4 \\ 3 & 5 & -7 \\ 1 & 6 & 2 \end{vmatrix} \begin{vmatrix} -2 & 1 \\ 3 & 5 \\ 1 & 6 \end{vmatrix} \begin{vmatrix} -2 & 1 \\ 3 & 5 \\ 1 & 6 \end{vmatrix}$$

$$= -2(5)(2) + 1(-7)(1) + 4(3)(6) - 4(5)(1) -$$

$$(-2)(-7)(6) - 1(3)(2)$$

$$= -20 - 7 + 72 - 20 - 84 - 6$$

$$= -65$$

12-

$$\begin{vmatrix} -1 & 1 & 2 \\ 3 & 0 & -5 \\ 1 & 7 & 2 \end{vmatrix}$$

$$\begin{vmatrix} -1 & 1 & 2 \\ 3 & 0 & -5 \\ 1 & 7 & 2 \end{vmatrix} \begin{vmatrix} -1 & 1 \\ 3 & 0 \\ 1 & 7 \end{vmatrix} \begin{vmatrix} -1 & 1 \\ 3 & 0 \\ 1 & 7 \end{vmatrix}$$

$$= -1(0)(2) + 1(-5)(1) + 2(3)(7) - 2(0)(1) -$$

$$(-1)(-5)(7) - 1(3)(2)$$

$$= 0 - 5 + 42 - 0 - 35 - 6 = -4$$

13-

$$\begin{vmatrix} 3 & 0 & 0 \\ 2 & -1 & 5 \\ 1 & 9 & -4 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 0 & 0 \\ 2 & -1 & 5 \\ 1 & 9 & -4 \end{vmatrix} \begin{vmatrix} 3 & 0 \\ 2 & -1 \\ 1 & 9 \end{vmatrix} \begin{vmatrix} 3 & 0 \\ 2 & -1 \\ 1 & 9 \end{vmatrix}$$

$$= 3(-1)(-4) + 0 + 0 - 3(5)(9) - 0 - 0$$

$$= +12 - 135 = -123$$

$$14 - \begin{bmatrix} c & -4 & 3 \\ 2 & 1 & c^2 \\ 4 & c-1 & 2 \end{bmatrix}$$

$$\begin{vmatrix} c & -4 & 3 \\ 2 & 1 & c^2 \\ 4 & c-1 & 2 \end{vmatrix}$$

$$= c(1)(2) + (-4)(c^2)(4) + 3(2)(c-1) -$$

$$3(1)(4) - c(c^2)(c-1) - (-4)(2)(2)$$

$$= 2c - 16c^2 + 6c - 6 - 12 - c^4 + c^3 + 16$$

$$= 8c - 16c^2 + c^3 - c^4 - 2$$

find λ for which $\det(A) = 0$

$$15 - A = \begin{bmatrix} \lambda-2 & 1 \\ -5 & \lambda+4 \end{bmatrix}$$

$$= (\lambda-2)(\lambda+4) - 1(-5)$$

$$= \lambda(\lambda+4) - 2(\lambda+4) + 5$$

$$= \lambda^2 + 4\lambda - 2\lambda - 8 + 5 = \lambda^2 + 2\lambda - 3$$

$$\lambda^2 + 2\lambda - 3 = 0$$

$$\lambda(\lambda+3) - 1(\lambda+3) = 0$$

$$(\lambda-1)(\lambda+3) = 0$$

$$\lambda = 1, \lambda = -3$$

Day: _____

16- $A = \begin{bmatrix} \lambda-4 & 0 & 0 \\ 0 & \lambda & 2 \\ 0 & 3 & \lambda-1 \end{bmatrix}$

$$\Rightarrow \begin{array}{c|cc|c} \lambda-4 & \lambda & 2 & -0 + 0 \\ \hline & 3 & \lambda-1 & \end{array}$$

$$\Rightarrow (\lambda-4)(\lambda^2 - \lambda - 6) = 0$$

$$\Rightarrow \lambda(\lambda^2 - \lambda - 6) - 4(\lambda^2 - \lambda - 6) = 0$$

$$\Rightarrow \lambda^3 - \lambda^2 - 6\lambda - 4\lambda^2 + 4\lambda + 24$$

$$\Rightarrow \lambda^3 - 5\lambda^2 - 2\lambda + 24$$

$$\Rightarrow \lambda - 4(\lambda^2 - 3\lambda + 2\lambda - 6) = 0$$

$$(\lambda-4)(\lambda-3)(\lambda+2) = 0$$

$$\lambda = 4, \lambda = 3, \lambda = -2$$

17- $A = \begin{bmatrix} \lambda-1 & 0 \\ 2 & \lambda+1 \end{bmatrix}$

$$\Rightarrow (\lambda-1)(\lambda+1) - 2(0)$$

$$(\lambda-1)(\lambda+1) = 0$$

$$\lambda = 1, \lambda = -1$$

Day: _____

$$18- A = \begin{bmatrix} \lambda-4 & 4 & 0 \\ -1 & \lambda & 0 \\ 0 & 0 & \lambda-5 \end{bmatrix}$$

$$\lambda-4 \mid \begin{array}{cc} \lambda & 0 \\ 0 & \lambda-5 \end{array} \mid \begin{array}{c} -4 \\ -1 \end{array} \mid \begin{array}{cc} 0 & 0 \\ 0 & \lambda-5 \end{array} \mid = 0$$

$$(\lambda-4)(\lambda^2-5\lambda) - 4(-\lambda+5) = 0$$

$$(\lambda-4)\lambda(\lambda-5) + 4\lambda - 20 = 0$$

$$\lambda = 4, \lambda = 0, \lambda = 5, \lambda = 5$$

$$(\lambda-4)(\lambda^2-5\lambda+4\lambda-20) = 0$$

$$(\lambda-4)(\lambda^2-\lambda-20)$$

$$(\lambda-4)(\lambda(\lambda-5)+4(\lambda-5)) = 0$$

$$(\lambda-4)(\lambda+4)(\lambda-5) = 0$$

$$\lambda(\lambda-4)(\lambda-5)+4(\lambda-5) = 0$$

$$(\lambda-5)(\lambda^2-4\lambda+4) = 0$$

$$(\lambda-5)(\lambda-2)(\lambda-2) = 0$$

$$\lambda = 5, \lambda = 2$$

9- Evaluate Determinant by cofactor Expansion

$$13- \begin{vmatrix} 3 & 0 & 0 \\ 2 & -1 & 5 \\ 1 & 9 & 4 \end{vmatrix} \quad \text{first row}$$

1- first row

$$\begin{vmatrix} 3 & 0 & 0 \\ 2 & -1 & 5 \\ 1 & 9 & 4 \end{vmatrix}$$

$$\rightarrow 3 \begin{vmatrix} -1 & 5 \\ 9 & 4 \end{vmatrix} - 0 + 0 = 3(-4 - 45) = -147$$

2- first ^{Third} column

$$\begin{vmatrix} 3 & 0 & 0 \\ 2 & -1 & 5 \\ 1 & 9 & 4 \end{vmatrix}$$

$$0 - 5 \begin{vmatrix} 3 & 0 \\ 1 & 9 \end{vmatrix} + 4 \begin{vmatrix} 3 & 0 \\ 2 & -1 \end{vmatrix}$$

$$= -5(27 - 0) + 4(-3 - 0)$$

$$= -135 - 12 = -147$$

3- Second Row

$$\begin{vmatrix} 3 & 0 & 0 \\ 2 & -1 & 5 \\ 1 & 9 & 4 \end{vmatrix}$$

$$\rightarrow -2 \begin{vmatrix} 0 & 0 \\ 9 & 4 \end{vmatrix} - 1 \begin{vmatrix} 3 & 0 \\ 1 & 4 \end{vmatrix} - 5 \begin{vmatrix} 3 & 0 \\ 1 & 9 \end{vmatrix}$$

$$= (0) - 1(12 - 0) - 5(27 - 0)$$

$$= -12 - 135 = -147$$

4- Second Column

$$\begin{vmatrix} 3 & 0 \\ 2 & -1 \\ 1 & -9 \end{vmatrix} \begin{vmatrix} 0 \\ 5 \\ 4 \end{vmatrix}$$

$$= 0 \cdot 1 \begin{vmatrix} 3 & 0 \\ 2 & 5 \end{vmatrix} - 9 \begin{vmatrix} 3 & 0 \\ 2 & 5 \end{vmatrix}$$

$$= -1(12 - 0) - 9(15 - 0)$$

$$= -12 - 135 = -147$$

5- Third Row

$$\begin{vmatrix} 3 & 0 & 0 \\ 2 & -1 & -5 \\ 1 & -9 & 4 \end{vmatrix}$$

$$= 1 \begin{vmatrix} 0 & 0 \\ -1 & -5 \end{vmatrix} + 9 \begin{vmatrix} 3 & 0 \\ 2 & 5 \end{vmatrix} + 4 \begin{vmatrix} 3 & 0 \\ 2 & -1 \end{vmatrix}$$

$$= 0 - 9(+15) + 4(-3)$$

$$= -135 - 12 = -147$$

6- ^{first} Third Column

$$\begin{vmatrix} 3 & 0 & 0 \\ 2 & -1 & 5 \\ 1 & 9 & 4 \end{vmatrix}$$

$$3 \begin{vmatrix} -1 & 5 \\ 9 & 4 \end{vmatrix} - 2 \begin{vmatrix} 0 & 0 \\ 9 & 4 \end{vmatrix} + 1 \begin{vmatrix} 0 & 0 \\ -1 & 5 \end{vmatrix}$$

$$3(-4 - 45) - 2(0) + 0 = -147$$

Question 20

$$\begin{vmatrix} -1 & 1 & 2 \\ 3 & 0 & -5 \\ 1 & 7 & 2 \end{vmatrix}$$

→ First Row

$$= \begin{vmatrix} -1 & 1 & 2 \\ 3 & 0 & -5 \\ 1 & 7 & 2 \end{vmatrix} \rightarrow -1 \begin{vmatrix} 0 & -5 \\ 7 & 2 \end{vmatrix} - 1 \begin{vmatrix} 3 & -5 \\ 1 & 2 \end{vmatrix} + 2 \begin{vmatrix} 3 & 0 \\ 1 & 7 \end{vmatrix}$$

$$= -1(0 + 35) - 1(6 + 5) + 2(21 - 0)$$

$$= -35 - 11 + 42 = -4$$

→ First Column

$$-1 \begin{vmatrix} 0 & -5 \\ 7 & 2 \end{vmatrix} - 3 \begin{vmatrix} 1 & 2 \\ 7 & 2 \end{vmatrix} + 1 \begin{vmatrix} 1 & 2 \\ 0 & -5 \end{vmatrix}$$

$$= -1(0 + 35) - 3(2 - 14) + 1(-5 + 0)$$

$$= -35 + 36 - 5 = -4$$

→ Second Row

$$-3 \begin{vmatrix} 1 & 2 \\ 7 & 2 \end{vmatrix} + 0 + 5 \begin{vmatrix} -1 & 1 \\ 1 & 7 \end{vmatrix}$$

$$= -3(2 - 14) + 0 + 5(-7 - 1)$$

$$= -3(-12) + 5(-8)$$

$$= 36 - 40 = -4$$

→ Second Column

$$\begin{array}{cc|cc|cc} -1 & 3 & -5 & +0 & +7 & -1 & 2 \\ 1 & 2 & & & & 3 & -5 \end{array}$$

$$= -1(6+5) - 7(5-6) = -1(11) - 7(-1)$$

$$= -11 + 7 = -4$$

→ Third Row

$$\begin{array}{cc|cc|cc|cc} 1 & 1 & 2 & -7 & -1 & 2 & +2 & -1 & 1 \\ 0 & -5 & & & 3 & -5 & & 3 & 0 \end{array}$$

$$1(-5-0) - 7(5-6) + 2(0-3)$$

$$= -5 + 7 - 6 = -5 + 1 = -4$$

→ Third Column

$$\begin{array}{cc|cc|cc|cc} 2 & +3 & 0 & +5 & -1 & 1 & +2 & -1 & 1 \\ 1 & 1 & 7 & & 1 & 7 & & 3 & 0 \end{array}$$

$$= 2(21) + 5(-7-1) + 2(0-3)$$

$$= 42 - 34 - 6 = -4$$

Evaluate determinant by co-factor expansion

$$21- A = \begin{bmatrix} -3 & 0 & 7 \\ 2 & 5 & 1 \\ -1 & 0 & 5 \end{bmatrix}$$

Expansion by 2nd column

$$= 0 + 5 \begin{vmatrix} -3 & 7 \\ -1 & 5 \end{vmatrix} - 0$$

$$= 5(-15 + 7) = 5(-8) = -40$$

$$22- A = \begin{bmatrix} 3 & 3 & 1 \\ -1 & 0 & -4 \\ 1 & -3 & 5 \end{bmatrix}$$

Expansion by 2nd row

$$= -1 \begin{vmatrix} 3 & 1 \\ -3 & 5 \end{vmatrix} + 0 + 4 \begin{vmatrix} 3 & 3 \\ 1 & -3 \end{vmatrix}$$

$$= -1(15 + 3) + 4(-9 - 3)$$

$$= -18 + 4(-12) = -18 - 48 = -66$$

$$23- A = \begin{bmatrix} 1 & k & k^2 \\ -1 & k & k^2 \\ +1 & k & k^2 \end{bmatrix}$$

Expansion by 1st column

$$1 \begin{vmatrix} k & k^2 \\ k & k^2 \end{vmatrix} - 1 \begin{vmatrix} k & k^2 \\ k & k^2 \end{vmatrix} + 1 \begin{vmatrix} k & k^2 \\ k & k^2 \end{vmatrix}$$

$$1(k^3 - k^3) = 0$$

Day:

$$24- A = \begin{bmatrix} k+1 & k-1 & 7 \\ 2 & k+3 & 4 \\ 5 & k+1 & k \end{bmatrix}$$

Expansion by 3rd column

$$7 \begin{vmatrix} 2 & k-3 \\ 5 & k+1 \end{vmatrix} - 4 \begin{vmatrix} k+1 & k-1 \\ 5 & k+1 \end{vmatrix} + k \begin{vmatrix} k+1 & k-1 \\ 2 & k-3 \end{vmatrix}$$

$$7(2k+2-5k+15) - 4(k^2+1+2k-5k+5) + k(k^2-2k-3-2k+2)$$

$$14k+14-35k+105-4k^2-4-8k+20k-20 +$$

$$k^3-2k^2-3k-2k^2+2k$$

$$-10k-8k^2+k^3+95$$

$$15- A = \begin{bmatrix} 3 & 3 & 0 & 5 \\ -2 & 2 & -0 & -2 \\ 4 & 1 & -3 & 0 \\ 2 & 10 & 3 & 2 \end{bmatrix}$$

Expansion by 2nd row

$$-2 \begin{vmatrix} 3 & 0 & 5 \\ 4 & 1 & -3 \\ 2 & 10 & 3 \end{vmatrix} + 2 \begin{vmatrix} 3 & 0 & 5 \\ 4 & -3 & 0 \\ 2 & 3 & 2 \end{vmatrix} + 0 \begin{vmatrix} 3 & 3 & 0 \\ 4 & 1 & -3 \\ 2 & 10 & 3 \end{vmatrix}$$

$$-2 \left[3 \begin{vmatrix} -3 & 0 \\ 3 & 2 \end{vmatrix} - 0 \begin{vmatrix} 1 & -3 \\ 10 & 3 \end{vmatrix} + 5 \begin{vmatrix} 1 & -3 \\ 10 & 3 \end{vmatrix} \right] + 2 \left[3 \begin{vmatrix} -3 & 0 \\ 3 & 2 \end{vmatrix} + 0 \begin{vmatrix} 4 & -3 \\ 2 & 3 \end{vmatrix} \right]$$

$$-2 \left[3 \begin{vmatrix} 1 & -3 \\ 10 & 3 \end{vmatrix} - 3 \begin{vmatrix} 4 & -3 \\ 2 & 3 \end{vmatrix} + 0 \right]$$

$$= -2 \left[3(-6-0) + 5(3+30) \right] + 2 \left[3(-6-0) + 5(12+6) \right] -$$

$$2 \left[3(3+30) - 3(12+6) \right]$$

$$= -2(-18 + 165) + 2(-18 + 90) - 2(99 - 54)$$

$$= -2(147) + 2(72) - 2(45)$$

$$= -294 + 144 - 90 = -240$$

26- $A = \begin{bmatrix} 4 & 0 & 0 & 1 & 0 \\ 3 & 3 & 3 & -1 & 0 \\ 1 & 2 & 4 & 2 & 3 \\ 9 & 4 & 6 & 2 & 3 \\ 2 & 2 & 4 & 2 & 3 \end{bmatrix}$

Expansion by Row 1

$$4 \begin{vmatrix} 3 & 3 & -1 & 0 \\ 2 & 4 & 2 & 3 \\ 4 & 6 & 2 & 3 \\ 2 & 4 & 2 & 3 \end{vmatrix} - 1 \begin{vmatrix} 3 & 3 & 3 & 0 \\ 1 & 2 & 4 & 3 \\ 9 & 4 & 6 & 3 \\ 2 & 2 & 4 & 3 \end{vmatrix}$$

Determinant(0) - 1

$$\begin{vmatrix} 3 & 3 & 3 & 0 \\ 1 & 2 & 4 & 3 \\ 9 & 4 & 6 & 3 \\ 2 & 2 & 4 & 3 \end{vmatrix}$$

$$-1 \left[3 \begin{vmatrix} 2 & 4 & 3 \\ 4 & 6 & 3 \\ 2 & 4 & 3 \end{vmatrix} - 3 \begin{vmatrix} 1 & 4 & 3 \\ 9 & 6 & 3 \\ 2 & 4 & 3 \end{vmatrix} + 3 \begin{vmatrix} 1 & 2 & 3 \\ 9 & 4 & 3 \\ 2 & 2 & 3 \end{vmatrix} \right]$$

$$= -1 \left[3(0) - 3 \left(\begin{vmatrix} 1 & 6 & 3 \\ 4 & 3 \end{vmatrix} - 4 \begin{vmatrix} 9 & 3 \\ 2 & 3 \end{vmatrix} + 3 \begin{vmatrix} 9 & 6 \\ 2 & 4 \end{vmatrix} \right) + \right.$$

$$\left. 3 \left(\begin{vmatrix} 1 & 4 & 3 \\ 2 & 3 \end{vmatrix} - 2 \begin{vmatrix} 9 & 3 \\ 2 & 3 \end{vmatrix} + 3 \begin{vmatrix} 9 & 4 \\ 2 & 2 \end{vmatrix} \right) \right]$$

$$\begin{aligned}
 &= -1 \left[-3 \left[(18-12) - 4(27-6) + 3(36-12) \right] + \right. \\
 &\quad \left. 3 \left[1(12-6) - 2(27-6) + 3(18-8) \right] \right] \\
 &= -1 \left[-3 \left[6 - 84 + 12 \right] + 3 \left[6 - 42 + 30 \right] \right] \\
 &= -1 \left[+18 + (-18) \right] \\
 &= -18 + 18 \\
 &= 0
 \end{aligned}$$

Determinant by Inspection

27- $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow (1)(-1)(1) = -1$

28- $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \rightarrow 2(2)(2) = 8$

29- $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 4 & 3 & 0 \\ 1 & 2 & 3 & 8 \end{bmatrix} \rightarrow (0)(2)(3)(8) = 0$

30- $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 6 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix} \rightarrow (1)(2)(3)(4) = 24$